

Part I

# SUMMARY of Applications of Integrals

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## Summary

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**SUMMARY: Velocity, Speed, Position, Displacement, Distance**

**Position**,  $s(t)$ , of an object at time  $t$ :  $s(t) = s(a) + \int_a^t v(z)dz$

**Displacement**,  $\Delta s = s(b) - s(a)$ ,  
of an object over the time interval  $[a, b]$ :  $s(b) - s(a) = \int_a^b v(t)dt$

**Distance** traveled by an object  
over the time interval  $[a, b]$ :  $\int_a^b |v(t)|dt$

**SUMMARY: Rate of Accumulation, Amount, Change in the Amount**

The amount,  $A(t)$ , of some  
substance/population at the time  $t$ :  $A(t) = A(a) + \int_a^t A'(z)dz$

The amount,  $A(b)$ ,  
over the time interval  $[a, b]$ :  $A(b) = A(a) + \int_a^b A'(t)dt$

The change in the amount,  $A(b) - A(a)$ ,  
over the time interval  $[a, b]$ :  $A(b) - A(a) = \int_a^b A'(t)dt$

**SUMMARY: Average Value, the Mean Value Theorem for Integrals**

**Average value**  $\bar{f}$ ,  
of the function  $f$  on the interval  $[a, b]$ :  $\bar{f} = \frac{1}{b-a} \int_a^b f(x)dx$

**Mean Value Theorem for Integrals**

## Summary

Let  $f$  be continuous on  $[a, b]$ . There exists a value  $c$  in  $(a, b)$  such that

$$\frac{1}{b-a} \int_a^b f(x) dx = f(c)$$

**Note 1:**  $f(c) = \bar{f}$ , the average value of  $f$  on  $[a, b]$ .

**Note 2:** The net area of the region between the curve  $y = f(x)$  and the  $x$ -axis is given by

$$\int_a^b f(x) dx = f(c)(b-a)$$

Part II

## Recitation Questions

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### Problem 1

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Problem 1

**Problem 1** Solve the following word problems:

- (a) The velocity function for an object moving along a line east/west is given by  $v(t) = -t^2 + 4t - 3$  feet per minute.
- (i) Find the total displacement the object traveled from 2 minutes to 6 minutes (assume east is positive).
  - (ii) Find the total distance the object traveled from 2 minutes to 6 minutes.
  - (iii) Suppose that the object's position 2 minutes into the trip is 5 feet of a placement marker. What is its position (relative to the placement marker) at 6 minutes.
- (b) Sammy the Snail sets up camp in the median of I-70 and, starting at noon and ending at 6pm, hikes back and forth along the highway. He starts his hike at his campsite. His velocity at time  $t$  hours (after noon) is given by  $v(t) = (t - 2)(t - 5)$  inches per hour. Find the total distance Sammy travelled on his hike.

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## Problem 2

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Problem 2

**Problem 2** Suppose that  $r(t) = r_0 e^{-kt}$  (with  $k > 0$ ) is the rate at which a nation extracts oil. The current rate of extraction is  $r(0) = 10^7$  barrels/yr. Also assume that the estimate of the total oil reserve (ie, the amount of oil remaining beneath the ground in this country) is  $2 \times 10^9$  barrels.

- (a) Find  $A(t)$ , the total amount of oil extracted by the nation after  $t$  years.
- (b) Evaluate  $\lim_{t \rightarrow \infty} A(t)$  and explain the meaning of this limit.
- (c) Find the minimum constant  $k$  for which the total oil reserves will last forever.
- (d) Suppose that the constant  $k$  is half the minimum value found in part (c). When will the nation deplete its oil reserve?

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# Problem 3

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Problem 3

**Problem 3** Assume that the rate of change (in dollars per day) of the price of shares of stock in the WeSaySo Company (with  $t$  in days) is modeled by the equation  $r(t) = -3t^2 + 30t - 63$  (note that this is technically a discrete function, but prices change so often with stocks that modeling this with a continuous function makes sense). Assume also that the price of a share of stock on day 1 (i.e.,  $t = 1$ ) is \$51. Answer the following questions:

- (a) Find the rate of change of price at  $t = 5$ .
- (b) Find the price of a share of stock at  $t = 5$ .
- (c) How fast is the rate of change of price changing at  $t = 5$ ?
- (d) How much did the price of a share of stock change in the first 6 days (i.e., on  $[1, 6]$ )?
- (e) What was the greatest rate of change of price during the first 6 days (i.e., on  $[1, 6]$ )?
- (f) What was the greatest price of a share of stock during the first 6 days (i.e., on  $[1, 6]$ )?

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# Problem 4

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*Problem 4*

**Problem 4** A cup of coffee has temperature  $20 + 75e^{-.02t}$  degrees (Celsius)  $t$  minutes after being poured into a cup. What is the average temperature of the coffee during the first half hour?

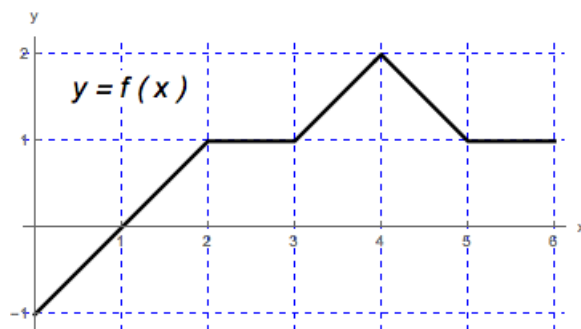
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# Problem 5

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**Problem 5** The graph of a function  $f$  defined on the interval  $[0, 6]$  is given in the figure.



- (a) Compute the net area of the region between the graph of  $f$  and the  $x$ -axis, on the interval  $[0, 6]$ .
- (b) Draw a rectangle with base on the  $x$ -axis,  $0 \leq x \leq 6$ , whose area is equal to the net area in part (a).
- (c) In the figure, mark a point  $c$  in  $(0, 6)$  such that  $f(c)$  is the height of the rectangle from part (b).
- (d) Using the figure and parts (a-c), what is the relationship between the rectangle from part (b), the net area from part (a), and the average value of  $f$  on  $[0, 6]$ ?

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# Problem 6

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*Problem 6*

**Problem 6** Find all points at which the given function equals its average value on the given interval.

(a)  $f(x) = e^x$        $[0, 4]$

(b)  $g(x) = \frac{\pi}{4} \sin(x)$        $[0, \pi]$

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# Problem 7

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*Problem 7*

**Problem 7** Find the average value of the function  $g(t) = 4te^{-t^2}$  on the interval  $[0, 3]$ .